ASHI'S

DESCRIPTIVE ANALYTICS REPORT

COLLEGE LIBRARY USAGE & STUDENT STUDY HABITS

Understanding library visits, study duration, resource usage, satisfaction, and service issues

DETAIL	INFORMATION
PROJECT TYPE	Descriptive Analytics Project
DATASET	300 synthetic library visit records
TOOLS	Microsoft Excel • Google Colab • Python • Pandas • Matplotlib
PROGRAM	BCA – Data Science and Artificial Intelligence
REPORT YEAR	2026

Academic project report • Synthetic data used for learning and demonstration

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Report note: This report uses a simulated dataset. Results are illustrative and should not be interpreted as actual statistics from a specific college library.

1. INTRODUCTION AND NOTE ON DATA

College libraries support students through study spaces, textbooks, reference material, computer access, and quiet environments for independent learning. Descriptive analytics can help summarize how students use these facilities and highlight patterns that may guide service improvements.

This report examines library usage and study habits across a sample of 300 simulated visit records during September 2026. The dataset includes visit purpose, student year, study area, visit time, study duration, resource type, books borrowed, internet issues, satisfaction rating, and book return status.

Note on data

Actual student-level library data may be private or unavailable for an academic exercise. Therefore, this project uses a synthetic dataset created for demonstration. The structure illustrates a descriptive analytics workflow and can be replaced with approved real data when available.

2. OBJECTIVES OF THE PROJECT

The main objectives are:

- To summarize library visit patterns and common study purposes.
- To compare average study duration across different purposes.
- To understand which study areas and resources are used most often.
- To examine student satisfaction and reported internet issues.
- To review book borrowing and return-status patterns.
- To present findings through charts, summary metrics, and practical recommendations.

Scope

The analysis is descriptive: it explains what appears in the dataset. It does not establish causes or prove that one factor directly creates another.

3. HOW THIS ANALYSIS WORKS AT REAL SCALE

A larger library analytics system can process approved data from library management systems, entry logs, surveys, and digital resource platforms. The same basic stages used in this project can be expanded to support more frequent reporting and larger datasets.

STAGE	PURPOSE	POSSIBLE TOOLS
Data collection	Collect visit, borrowing, resource, and feedback records.	Library system exports, Excel
Data preparation	Remove duplicates, standardize categories, check missing values and dates.	Excel, Python, Pandas
Descriptive analysis	Calculate counts, averages, percentages, and grouped summaries.	Pandas, Excel
Visualization	Present patterns in charts and tables for easier interpretation.	Matplotlib, Excel charts
Reporting	Share findings with librarians and academic administrators.	PDF reports, dashboards

Potential real-world applications

- Plan quiet and group study spaces according to usage.
- Identify busy periods to improve staffing and seating availability.
- Review internet reliability when students report service interruptions.
- Understand demand for textbooks, reference books, and e-resources.
- Monitor borrowing and return records to support timely reminders.

Privacy consideration

In a real deployment, student identifiers should be removed or protected, access should be limited to authorized staff, and any use of survey or attendance data should follow institutional rules.

4. DATA COLLECTION, PREPARATION & ANALYSIS

The dataset was generated with 300 synthetic library visits dated across September 2026. Each row represents one visit record rather than one unique student. A student could therefore appear in multiple visits in a real version of this project.

Fields included in the dataset

FIELD	DESCRIPTION
Visit Date	Date on which the visit occurred.
Student Year	Academic year category: 1st, 2nd, or 3rd year.
Visit Purpose	Exam preparation, individual study, group study, or research.
Study Area	Quiet zone, reading hall, group room, or computer section.
Visit Time	Morning, afternoon, or evening.
Study Duration	Estimated duration of the visit in hours.
Resource Type	Textbooks, reference books, e-resources, or mixed resources.
Books Borrowed	Number of books borrowed during the visit.
Internet Issues	Whether an internet issue was reported.
Satisfaction Rating	Illustrative rating from 1 (low) to 5 (high).
Book Return Status	Returned, not returned, or not applicable when no book was borrowed.

Preparation steps

1. Created consistent category labels for visit purpose, area, and resource type.
2. Checked that study duration values were positive and ratings stayed within the 1–5 range.
3. Separated records with borrowed books before calculating the return percentage.
4. Grouped records by category to calculate counts, averages, and percentages.
5. Generated charts to make patterns easier to compare.

5. KEY PERFORMANCE INDICATORS (KPIs)

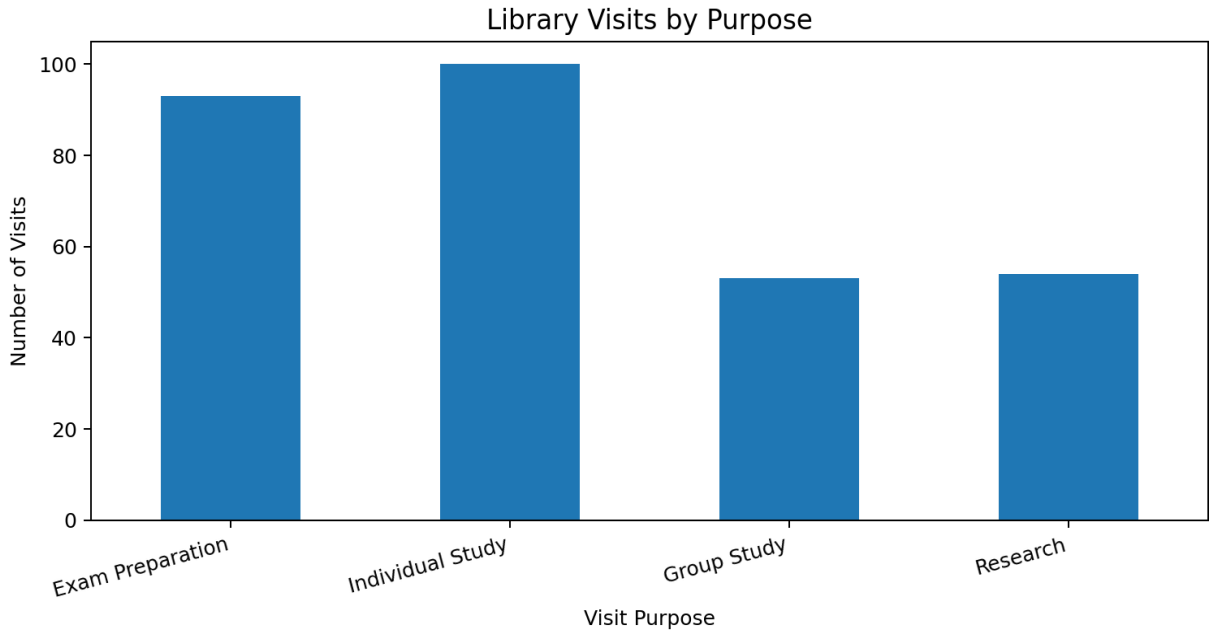
The following metrics summarize the synthetic library dataset. They provide a quick overview before examining individual charts.

METRIC	VALUE	INTERPRETATION
Total visit records	300	Number of records included in the analysis.
Total study time	739.9 hours	Combined duration across all recorded visits.
Average duration per visit	2.47 hours	Mean study time for one visit record.
Average satisfaction	3.92 / 5	Mean of the illustrative satisfaction ratings.
Visits reporting internet issues	22.7%	Share of records marked as having an issue.
Average books borrowed	1.31	Mean number of books borrowed per visit record.
Return rate among borrowed records	77.3%	Returned records divided by records where at least one book was borrowed.
Most common visit purpose	Individual Study	Purpose with the highest number of records.
Most used study area	Quiet Zone	Study area with the highest number of records.

Important: These values are calculated from the generated synthetic dataset and are included to demonstrate KPI calculation. They are not official college library statistics.

6. LIBRARY VISITS BY PURPOSE

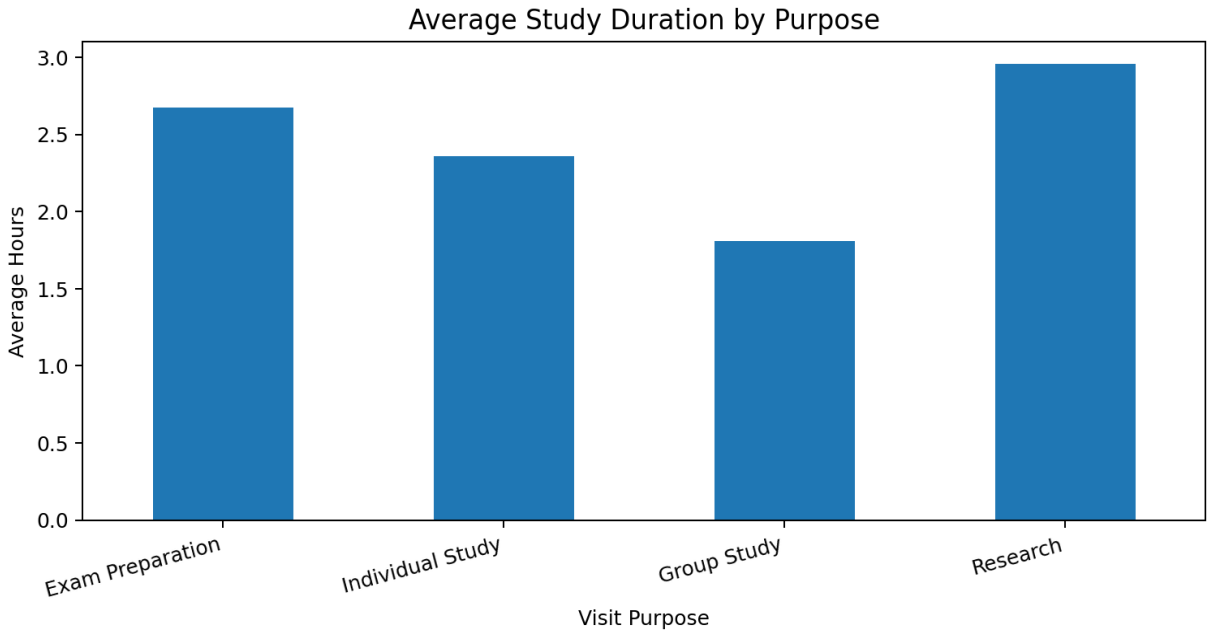
This chart compares the number of visits recorded for exam preparation, individual study, group study, and research. It helps show the main reasons students use library facilities in the sample.



Observation: Individual Study is the most frequently recorded visit purpose in this dataset. Library staff could use purpose-based counts as one input when reviewing space and resource needs.

7. AVERAGE STUDY DURATION BY PURPOSE

Average duration helps compare how long visit records last for different study activities. Averages are descriptive summaries and do not indicate how effective or productive a session was.

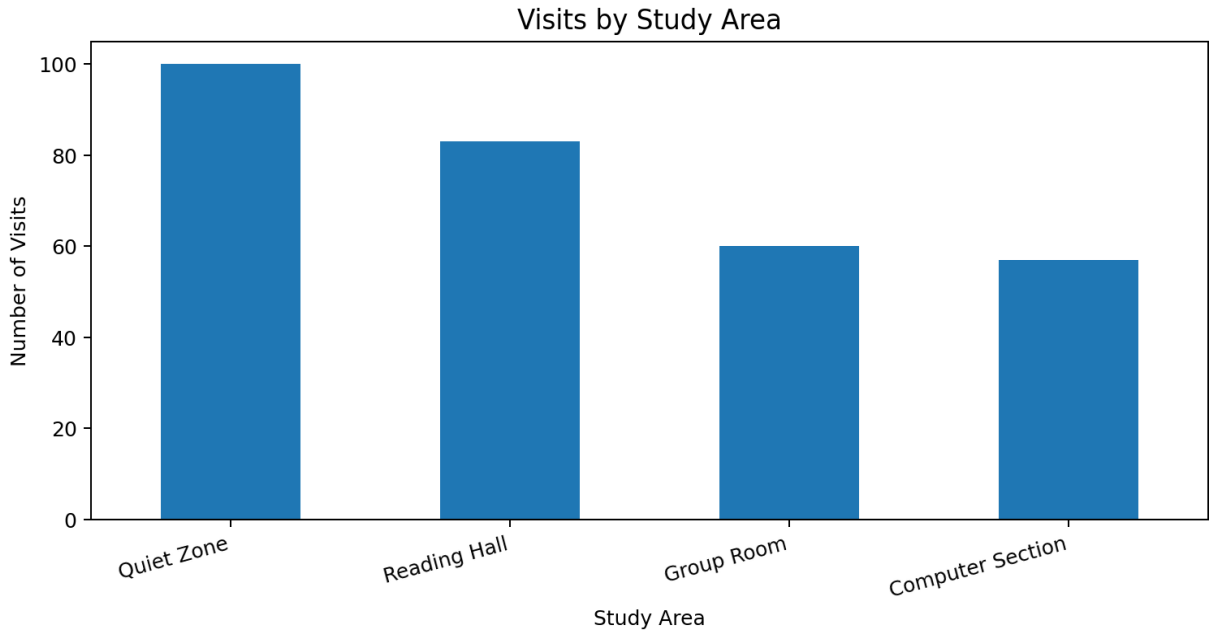


VISIT PURPOSE	AVERAGE DURATION (HOURS)
Exam Preparation	2.68
Individual Study	2.36
Group Study	1.81
Research	2.96

Observation: Research and exam-preparation visits have been assigned longer typical durations in the synthetic data than group-study visits. This pattern is illustrative, not a measured conclusion about actual students.

8. VISITS BY STUDY AREA

Study-area counts show how recorded visits are distributed across the quiet zone, reading hall, group room, and computer section.



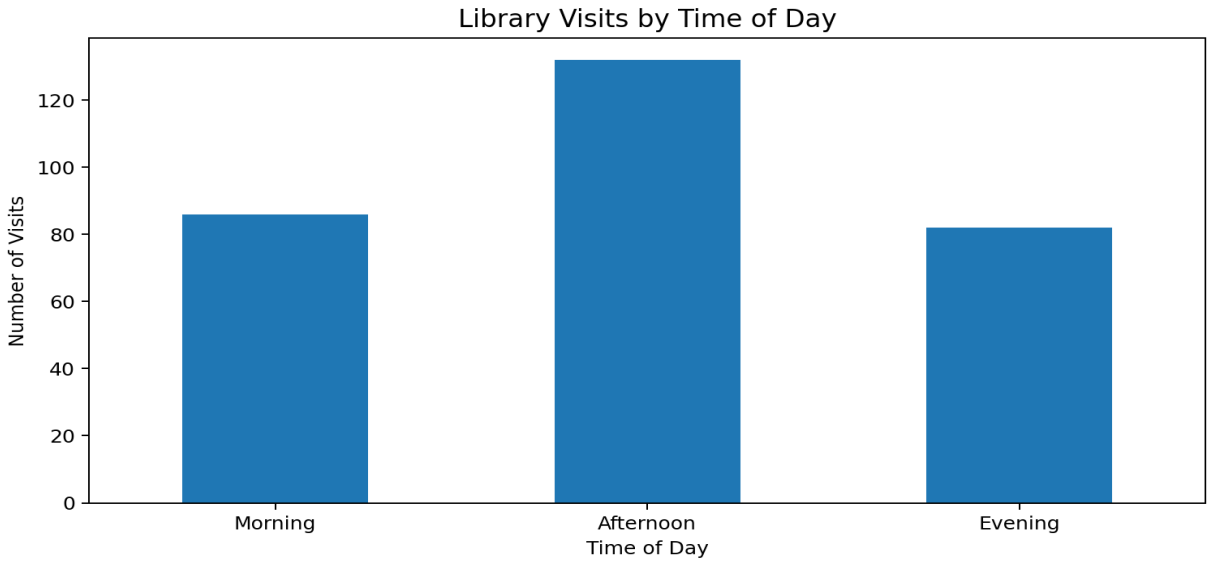
STUDY AREA	VISITS	SHARE
Quiet Zone	100	33.3%
Reading Hall	83	27.7%
Group Room	60	20.0%
Computer Section	57	19.0%

Observation: Quiet Zone is the most frequently used area in the sample. Actual space planning should also consider occupancy levels, seat availability, and the length of each visit.

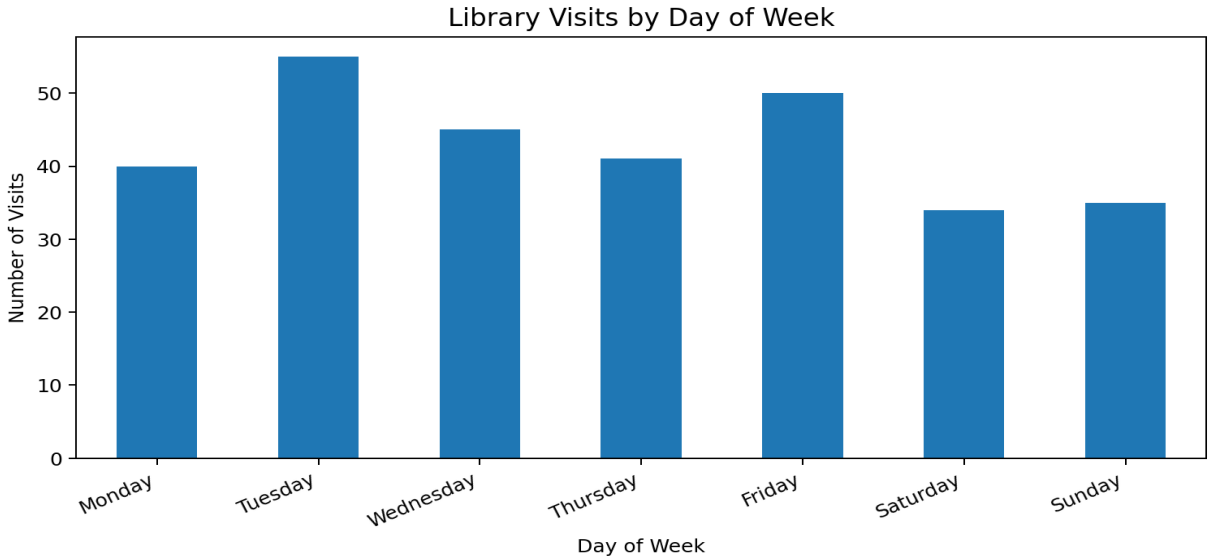
9. VISITS BY TIME OF DAY AND WEEKDAY

Time-of-day and weekday summaries help identify when library use is more common. These charts can support questions about opening hours, seating demand, and staffing.

Visits by time of day



Visits by weekday

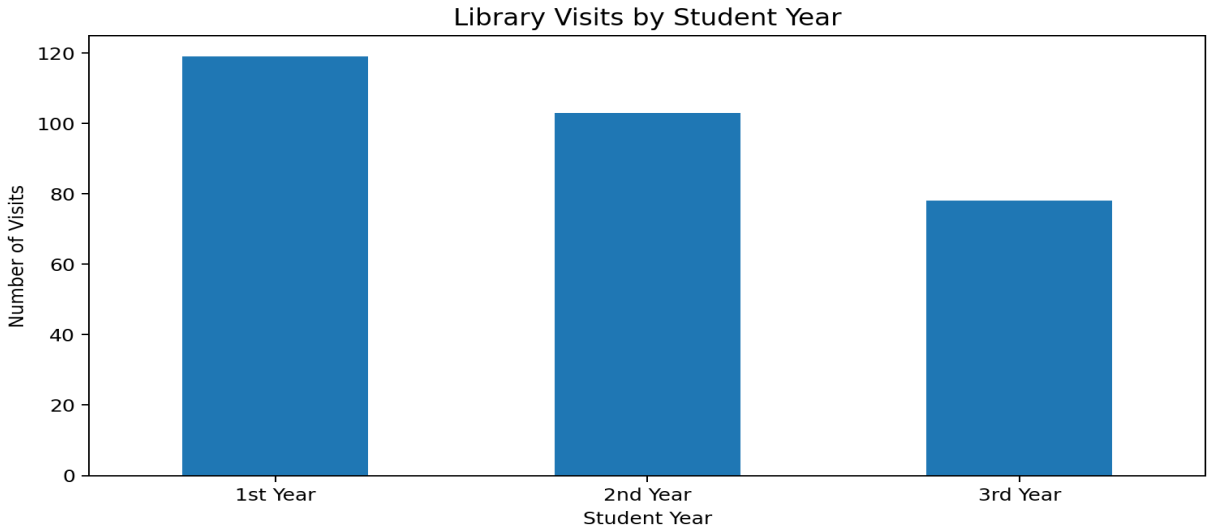


Observation: Afternoon has the highest count among the three time categories in this sample. Weekday patterns are based on the generated visit dates and should be validated against real attendance records before operational decisions.

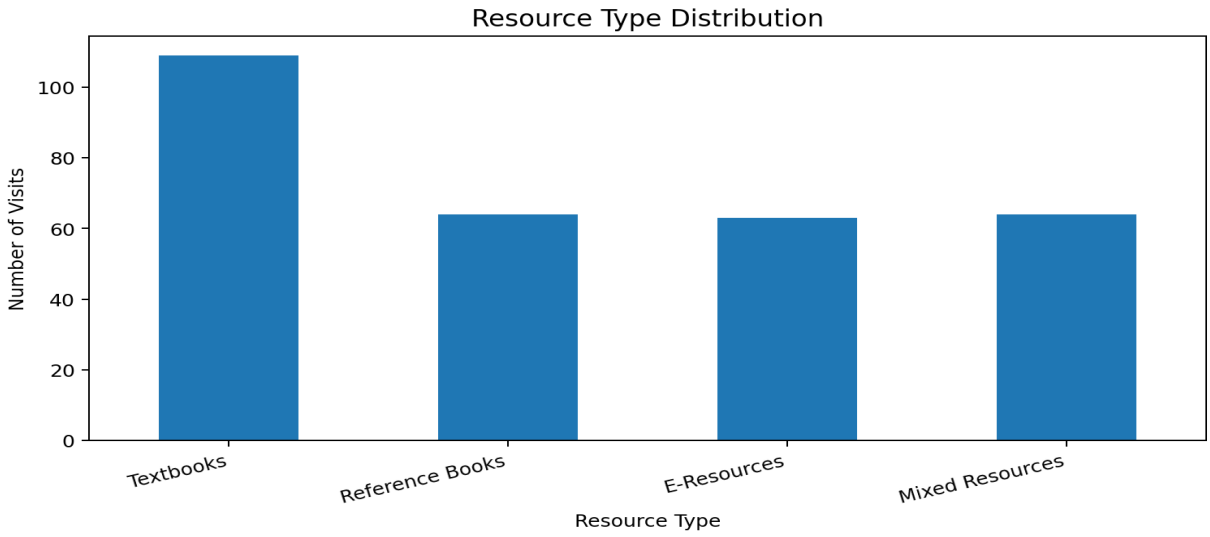
10. STUDENT YEAR AND RESOURCE DISTRIBUTION

This section compares the number of visit records by academic year and summarizes the resource types recorded during visits.

Visits by student year



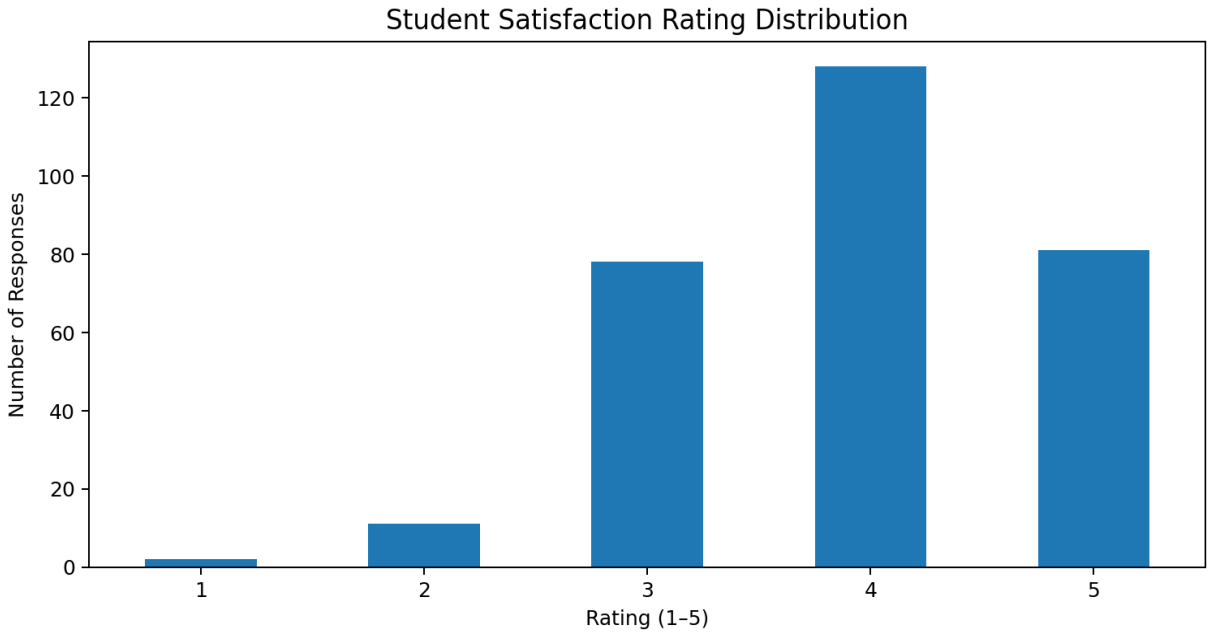
Resource type distribution



Observation: The distributions show which year groups and resource categories appear most often in the sample. In real data, these results could help inform book acquisition, digital-resource access, and student support planning.

11. STUDENT SATISFACTION ANALYSIS

Satisfaction ratings provide a simple summary of how users evaluate their library experience. In this synthetic dataset, ratings range from 1 (low satisfaction) to 5 (high satisfaction).



RATING	NUMBER OF RECORDS	SHARE
1	2	0.7%
2	11	3.7%
3	78	26.0%
4	128	42.7%
5	81	27.0%

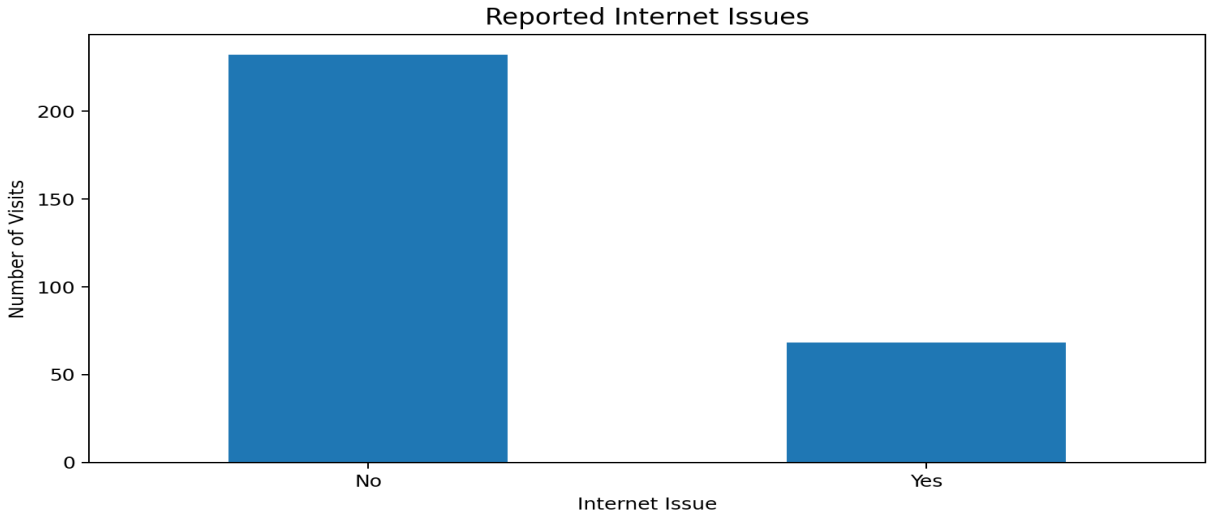
Average satisfaction rating: 3.92 out of 5.

Interpretation: Ratings can help identify whether the overall experience appears positive or whether feedback needs closer review. A real study should include the survey question, response count, and collection method.

12. INTERNET ISSUES AND BOOK RETURNS

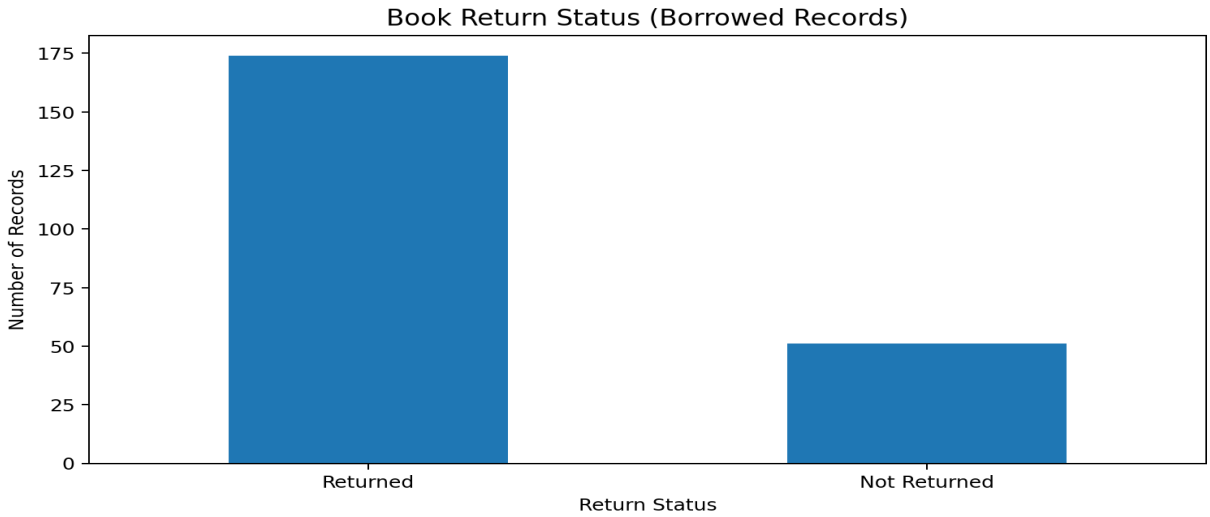
This section reviews two operational indicators: whether internet issues were reported and whether borrowed books were marked as returned.

Reported internet issues



Internet issues were marked in 22.7% of visit records. In an actual library, repeated issue reports could be compared with support tickets or network availability logs.

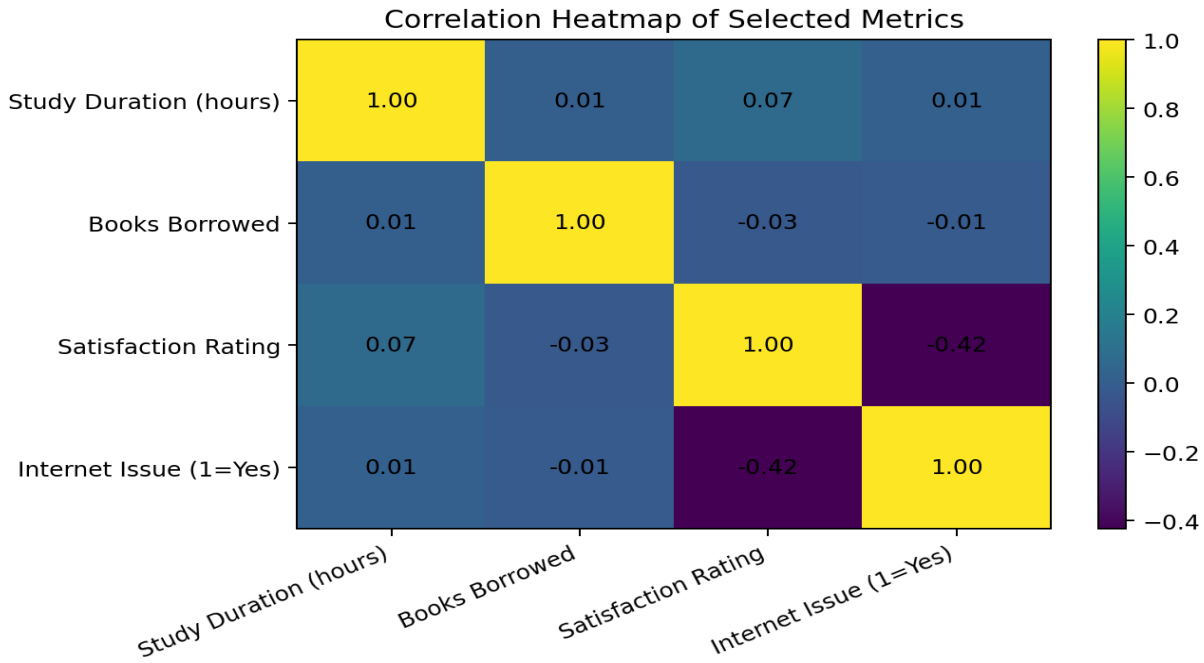
Book return status



Among records where at least one book was borrowed, the return rate is 77.3%. Records with zero borrowed books are excluded from this calculation.

13. CORRELATION ANALYSIS

A correlation matrix summarizes the direction and strength of linear relationships among selected numerical variables. Here, study duration, books borrowed, satisfaction rating, and an encoded internet-issue indicator are compared.



How to read the heatmap

Values closer to +1 indicate a positive linear relationship; values closer to –1 indicate a negative linear relationship; values around 0 indicate a weak linear relationship. Correlation alone does not establish cause and effect.

Because this is synthetic data, any relationships shown are only illustrative. Real-world conclusions would require actual records, data-quality checks, and careful interpretation.

14. SUMMARY OF FINDINGS AND RECOMMENDATIONS

Summary of key metrics

METRIC	VALUE
Total visit records	300
Total study duration	739.9 hours
Average study duration	2.47 hours per visit
Average satisfaction	3.92 / 5
Internet issue records	22.7%
Book return rate among borrowed records	77.3%
Most common purpose	Individual Study
Most used study area	Quiet Zone

Main observations

• Individual Study is the most common visit purpose in the sample.
• Quiet Zone is the most frequently recorded study area.
• Study duration varies by visit purpose, making it useful to compare average hours alongside visit counts.
• Satisfaction ratings, internet reports, and book return records provide additional service-quality indicators.
• Time-of-day and weekday summaries can help frame questions about library demand.

Recommendations for a real library

1. Review seating and room use during the busiest periods.
2. Track internet complaints consistently and record when issues occur.
3. Monitor overdue books using the library's official borrowing records.
4. Collect feedback with a consistent, anonymous satisfaction survey.
5. Repeat the analysis monthly to identify changes over time.

15. NEXT STEPS: FROM DESCRIPTIVE TO PREDICTIVE

This report describes what appears in the library-usage dataset. A future predictive analytics stage could use historical, approved records to estimate what may happen next, provided the dataset is sufficiently large and reliable.

Possible future extensions

• Visit forecasting: estimate library demand by day or time of day.
• Seat-demand prediction: estimate which study areas may become busy.
• Resource-demand analysis: forecast demand for selected books or e-resources.
• Satisfaction analysis: examine whether reported service issues are associated with lower ratings.
• Overdue-return risk: identify borrowing records that may need reminders, subject to appropriate privacy safeguards.

Suggested workflow

STEP	ACTIVITY
1. Collect	Use approved historical visit, borrowing, and feedback data.
2. Prepare	Clean missing values, inconsistent labels, duplicates, and dates.
3. Explore	Compare patterns across months, terms, weekdays, and areas.
4. Model	Test suitable forecasting or classification methods on historical data.
5. Evaluate	Compare predictions with actual outcomes and report model limitations.

Final conclusion

This project demonstrates how descriptive analytics can turn basic library records into understandable metrics and charts. The findings are illustrative because the dataset is synthetic, but the workflow provides a practical starting point for future data-driven library planning.

Tools used: Microsoft Excel, Google Colab, Python, Pandas, Matplotlib.